

Jointly-Learned Boundary-Risk Posterior for Wind Plant Reserve Pricing under Telemetry Degradation: A Defense-in-Depth Approach

Junyu Li and Juntao Du* School of Statistics and Applied Mathematics, Anhui University of Finance and Economics, Bengbu 233030, China

*Corresponding author: dujuntao@aufe.edu.cn

Abstract—Reserve sizing near the transition from maximum power point tracking (MPPT) to blade-pitch regulation requires high-fidelity awareness of turbine operating boundaries. However, industrial wind-plant Supervisory Control and Data Acquisition (SCADA) telemetry streams are inherently vulnerable to communication packet loss, multi-step transmission latency, sensor noise, and uncalibrated or unobservable pitch channels (particularly at third-party aggregator or Virtual Power Plant interfaces). We establish that deterministic physical rules (such as continuous soft-pitch rules based on aerodynamic power curves) exhibit remarkable economic efficiency under pristine telemetry (achieving the lowest baseline reserve cost of 871,408 kW·h at 1-hour dispatch lead time $h = 6$), yet suffer catastrophic brittleness under communication delays or noise, where shortage violation rates surge to $12.52\% \pm 1.82\%$ ($h = 6$) and $12.60\% \pm 1.94\%$ ($h = 1$), severely breaching the 10% grid reliability compliance limit. Conversely, shallow tree baselines (Missingness-Aware GBDT) capitalize on 10-minute lag autocorrelation at $h = 1$ but collapse across dispatch-grade horizons ($h = 6$), inflating reserve costs by 13.44% to 29.15% ($p < 0.0001$) due to their structural inability to capture spatial wake physics. To resolve this dilemma, we propose a synergistic defense-in-depth architecture combining high-fidelity physical aerodynamic priors with a jointly-learned spatio-temporal dynamic graph neural network that serves as an autonomous blind-spot fallback. When primary telemetry degrades, the deep graph representation dynamically reconstructs operating risk from cross-sensor electromechanical signatures (dynamic wake graph geometry, active power transients, voltage/reactive dynamics) when primary pitch readings are delayed or unobservable. Across a 5-seed benchmark on the 134-turbine WTB plant, we show that nominal pricing efficiency is driven by end-to-end task-loss alignment (dense and routed heads achieve statistical parity under clean telemetry, 95% CI $[-1,148, +65,456]$ kW·h), while dynamic MoE routing localizes its distinct empirical value to fault isolation under continuous telemetry impairment, saving +16,267 kW·h ($p = 0.0251$) under 6-step delay and +21,006 kW·h ($p < 0.05$) under sensor noise against unrouted dense representations, while outperforming global quantiles by 17,010 to 39,962 kW·h and avoiding the severe tail conservatism of uncalibrated pinball regression (690,695 kW·h savings over Frozen Backbone Direct Quantile MLP). Under Markov-Gilbert burst drops, it sustains 0.994 recall and 0.967 F1 (+0.064 F1 gain over the rule); under a six-step delay, it sustains 0.933 recall (+0.737 gain). At the Point of Common Coupling (PCC) bus, fleet-wide spatial portfolio smoothing across active turbine subsets (averaging ~ 121 active turbines) preserves an economic savings of -11.22 M kWh (95% CI $[-21.55\text{M}, -2.85\text{M}]$) over global PCC baselines. Replicated across commercial UK wind plants with turbine-calibrated rated wind speeds (Kelmars MM92 at 12.5 m/s, Penmanshiel MM82 at 14.5 m/s), consequence signatures persist significantly above chance under counterfactual withheld-channel stress testing, and decadal walk-forward rolling recalibration saves 0.804M kWh/year across 17.6 operating years (12 of 12 clean folds saving, $p = 0.00024$; 12 of 13 full folds, $p = 0.00171$). The resulting framework delivers an accountable,

physics-data synergistic reserve pricing layer for digitalized power grids.

Index Terms—Wind plant reserve pricing; Point of Common Coupling (PCC); telemetry degradation; defense-in-depth; high-fidelity physical prior; spatio-temporal dynamic graph; blind-spot fallback; MPPT-to-pitch transition.

I. INTRODUCTION

Wind-farm reserve screening near the transition from maximum power point tracking (MPPT) to blade-pitch control requires precise identification of the active turbine operating regime. This operational boundary matters acutely because wind-power forecasting is economically coupled to ramp exposure, operating reserve sizing, and non-convex imbalance shortfall penalties rather than to aggregate RMSE alone [1], [2], [3]. In the MPPT aerodynamic region (Region 2), active power responds cubically to wind-speed variations. Once blade-pitch control becomes active near rated wind speed (Region 3), aerodynamic rotor efficiency is actively curtailed to clamp generation at nameplate capacity. A small unpredicted wind-speed deviation near rated operation can therefore thrust a turbine across a control cliff, generating severe power shortfall penalties even when fleet-average forecast errors appear benign.

In industrial wind plant operations, however, continuous and pristine SCADA telemetry streams cannot be guaranteed. Telemetry channels across distributed turbine networks are routinely afflicted by communication packet dropouts, multi-step transmission latency (e.g., 10 to 60 minutes), asynchronous timestamp misalignment, and sensor calibration noise [4]. In the commercial Kelmars wind plant, for example, 99.6% of turbine status events carry second-level timestamps that fall strictly between 10-minute SCADA grid points, confirming that confirming-stream transmission delay and asynchrony are endemic structural features of distributed SCADA networks rather than rare edge cases. Furthermore, in non-intrusive aggregator, Virtual Power Plant (VPP), or Transmission System Operator (TSO) pre-dispatch triage settings, internal turbine blade-pitch registers are frequently unobservable due to commercial boundaries, proprietary OEM protocol firewalls, or legacy substation gateways. When grid operators must clear reserve margins before real-time delivery, relying on delayed or unobservable confirming telemetry poses severe reliability risks.

Prior engineering practice and research have largely treated physical models and data-driven forecasters as mutually ex-

clusive alternatives. On the one hand, high-fidelity physical priors—such as continuous soft-pitch rules based on manufacturer power curves—exhibit exceptional economic efficiency under ideal, clean observations, achieving the lowest baseline reserve cost (871,408 kW · h at 1-hour dispatch lead time $h = 6$). However, pure physical rules exhibit **catastrophic brittleness**: when SCADA telemetry experiences communication delays or sensor noise, physical threshold logic misclassifies the turbine’s control state, causing reserve shortage violation rates to explode to $12.52\% \pm 1.82\%$ ($h = 6$) and $12.60\% \pm 1.94\%$ ($h = 1$), severely breaching the strict 10% grid reliability compliance limit. On the other hand, shallow machine-learning baselines like Missingness-Aware GBDT capitalize on 10-minute lag wind-speed autocorrelation at ultra-short horizons ($h = 1$), but undergo a catastrophic breakdown at dispatch-grade horizons ($h = 6$), inflating reserve costs by 13.44% to 29.15% ($p < 0.0001$) because decision trees cannot represent spatial wake propagation across turbine arrays. Direct uncalibrated two-stage architectures (Frozen Backbone Direct Quantile MLP) suffer severe tail conservatism, over-estimating reserve margins by 44.57% to 72.02% ($p < 0.01$) by hoarding excessive reserves under direct pinball regression.

To resolve this fundamental tension, we propose a synergistic defense-in-depth framework: **“High-Fidelity Physical Prior + Spatio-Temporal Dynamic Graph Blind-Spot Fallback”**. Under nominal operating conditions with intact telemetry, high-fidelity physical aerodynamic priors govern primary reserve allocation, ensuring maximum economic efficiency. When SCADA telemetry enters blind spots (packet dropouts, transmission delays, uncalibrated pitch channels), an end-to-end spatio-temporal directed-diffusion graph neural network coupled with an adaptive Mixture-of-Experts (MoE) boundary-router serves as an autonomous safety net. By propagating dynamic wake geometry and cross-sensor electromechanical signatures (active power transients, terminal voltage, and reactive dynamics), the learned posterior maintains sharp, grid-compliant risk awareness even when primary blade-pitch telemetry is delayed, unobservable across aggregator boundaries, or withheld under counterfactual stress testing. Furthermore, we verify that this boundary-risk posterior survives the wind farm spatial portfolio smoothing effect at the Point of Common Coupling (PCC) bus ($P_{\text{farm}} = \sum_{i \in \mathcal{G}_t} P_{i,t}$, averaging ~ 121 active turbines), delivering verifiable economic savings after fleet-wide error cancellation.

This paper makes four bounded, verifiable contributions: 1. **Synergistic Defense-in-Depth Architecture**: We formalize a collaborative framework uniting high-fidelity aerodynamic physical priors with a spatio-temporal dynamic graph blind-spot fallback, saving -11.22M kWh (95% CI $[-21.55\text{M}, -2.85\text{M}]$) over global PCC quantiles and -14.94M kWh over Gaussian baselines at the Point of Common Coupling (PCC) bus. 2. **Multi-Horizon 5-Seed Dispatch Benchmark**: Across 5 seeds (201–205) and four degradation regimes, nominal pricing efficiency is driven by end-to-end task-loss alignment (dense and routed heads achieve statistical parity in Clean, 95% CI $[-1,148, +65,456]$ kW · h), while dynamic MoE routing localizes its distinct value to continuous telemetry impairment ($p = 0.0251$ under Delay-6, $p < 0.05$ under noise), outperforming GBDT ($p < 0.0001$) and avoiding

direct pinball conservatism ($p < 0.01$). 3. **Physical Brittleness & Degradation Resilience**: We expose the breakdown of deterministic rules under delay ($>12.5\%$ violations) and prove the deep graph fallback maintains compliance ($<8.1\%$ violation rate) while sustaining 0.994 recall and 0.967 F1 under Markov-Gilbert burst drops ($+0.064$ gain) and 0.933 recall under 6-step delays ($+0.737$ gain). 4. **Multi-Farm Generalization & Decadal Walk-Forward Durability**: We replicate the framework across commercial wind plants under heterogeneous turbine types (ENGIE La Haute Borne, Kelmarsh MM92 with $v_{\text{rated}} = 12.5$ m/s, Penmanshiel MM82 with $v_{\text{rated}} = 14.5$ m/s), proving consequence signatures persist under counterfactual withheld-channel testing, and show rolling recalibration under frozen backbones yields cost savings across 12 of 12 clean folds ($p = 0.00024$; 12 of 13 full folds, $p = 0.00171$) over 17.6 cumulative operating years.

II. RELATED WORK

A. From average wind-power accuracy to transition-window risk

Wind-power forecasting has progressed from site-specific statistical models to spatio-temporal architectures that represent ramps, uncertainty and spatial coupling [1], [3]. Graph models provide a strong accuracy reference by propagating information across the network [5], [6], [7]. Wind-specific studies have further introduced wake-aware graphs, SCADA integration and physics-guided constraints [8], [9], [10], [11]. These advances improve prediction, but they do not by themselves expose the active turbine-control law to a reserve planner.

MoE routing can represent local response laws, but prediction loss alone does not determine whether the resulting partition has operating meaning [12], [13]. Physics-guided machine learning supplies general principles for introducing scientific constraints, while wind-turbine operating states remain limited by SCADA observability [4], [14], [15]. The MPPT-to-pitch transition itself is a documented control-engineering structure: variable-speed turbines maximise capture below rated wind (Region 2), and pitch regulation takes over above it [16], [17], [18], [19]. Our distinction is therefore the location of the constraint. Earlier work constrains predictions or lets prediction loss organise experts; we constrain the route against a declared operating boundary and audit that assignment explicitly. This framing makes no anchor-free discovery claim. It asks whether route provenance can support a bounded reserve diagnostic.

Forecast value is ultimately realised through reserve and commitment decisions. Previous studies show that variable generation changes reserve requirements, while probabilistic and quantile forecasts provide a bridge from prediction error to decision risk [2], [20], [21], [22], [23]. We address an earlier link in that chain: whether the transition window can be identified before the confirming label is available. The reserve audit is therefore a pre-dispatch screening log for balancing and imbalance-risk triage, not a probabilistic dispatch or market-clearing model.

III. PHYSICS-INFORMED FRAMEWORK FOR OPERATIONAL BOUNDARY IDENTIFICATION

A. Problem setup and notation

The forecasting task is written to separate two decisions that dense models often merge: predicting the future and deciding which local mapping should be active at the anchor time.

This separation follows the spatio-temporal graph forecasting formulation in which a graph encoder maps a history window to a future trajectory [5], [24]. Let $G = (V, E)$ denote a spatial graph with N nodes. For each node $i \in V$ and time step t , we observe a feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^F$ and predict a scalar target $y_{i,t} \in \mathbb{R}$. Given a history window of length H and a prediction horizon of length P , the forecasting task is

$$\hat{\mathbf{Y}}_{t+1:t+P} = \mathcal{F}(\mathbf{X}_{t-H+1:t}, \mathcal{A}_{t-H+1:t}),$$

where $\mathbf{X}_{t-H+1:t} \in \mathbb{R}^{H \times N \times F}$ and $\mathcal{A}_{t-H+1:t}$ denotes either a time-varying directed graph sequence (WTB) or a static graph repeated over time (ERA5). Invalid or missing targets are excluded by a supervision mask.

The information boundary is strict. All inputs, graph weights, regime anchors, and gate anchors are observed no later than the forecast issue time t ; supervised targets begin at $t + 1$. In WTB, the current active-power channel $\text{Patv}_{i,t}$ is treated as an issue-time status input, whereas future active power is used only as the prediction target and validity mask. This keeps the multi-step horizon free of target leakage while leaving the no-Patv and lagged-Patv variants as deployment limitations.

Routing is node-level: each node at each anchor time receives its own gate distribution $\mathbf{g}_{i,t}$. This matters because nearby turbines or grid cells can occupy different local regimes within the same sequence. Core symbols are defined where they first appear, and auxiliary loss definitions are provided in Appendix A.

B. Operating-boundary anchors and high-fidelity physical prior (Tier 1)

The gate is not supervised everywhere. It is anchored where the physical aerodynamic interpretation is clearest, while ambiguous samples remain governed by prediction loss and routing regularisation. This design embodies the first tier of our defense-in-depth posture: **a high-fidelity physical prior** calibrated to the manufacturer specifications of each turbine model [4], [14], [15].

Aerodynamic operating regimes and turbine-calibrated anchors: In wind turbine control engineering, the transition from Maximum Power Point Tracking (Region 2) to active blade-pitch regulation (Region 3) is governed by the turbine's aerodynamic power-coefficient curve $C_p(\lambda, \theta)$ and its rated wind speed u_{rated} . Below rated wind speed, blades maintain a minimum pitch angle to maximise aerodynamic capture; above rated wind speed, the pitch actuator rotates the blades to shed excess aerodynamic power and protect the generator.

Let $\bar{p}_{i,t} = \frac{1}{3} \sum_{k=1}^3 p_{i,t}^{(k)}$ denote the average blade pitch angle across all three blades for turbine i at dispatch anchor time t . Writing $w_{i,t} = \text{Wspd}_{i,t}$ for anemometer wind speed, the declared operating boundary is defined with turbine-calibrated cut-in threshold u_{idle} , rated wind speed u_{rated} , and pitch-activation angle p_{th} :

$$R_{i,t} = \begin{cases} 0, & w_{i,t} < u_{\text{idle}} \quad (\text{idle}), \\ 1, & u_{\text{idle}} \leq w_{i,t} \leq u_{\text{rated}}, \bar{p}_{i,t} < p_{\text{th}} \quad (\text{MPPT}), \\ 2, & w_{i,t} > u_{\text{rated}}, \bar{p}_{i,t} \geq p_{\text{th}} \quad (\text{pitch-control}), \\ 3, & \text{otherwise} \quad (\text{transitional/ambiguous}). \end{cases}$$

Crucially, aerodynamic rated wind speeds are calibrated to the specific turbine technology at each wind plant: WTB (134 turbines: $u_{\text{idle}} = 3.0 \text{ m s}^{-1}$, $u_{\text{rated}} = 10.5 \text{ m s}^{-1}$, $p_{\text{th}} = 2.0^\circ$), Kelmarsh (6 MM92 turbines: $u_{\text{idle}} = 3.0$, $u_{\text{rated}} = 12.5 \text{ m s}^{-1}$, $p_{\text{th}} = 1.0^\circ$), Penmanshiel (15 MM82 turbines: $u_{\text{idle}} = 3.0$, $u_{\text{rated}} = 14.5 \text{ m s}^{-1}$, $p_{\text{th}} = 1.0^\circ$), and ENGIE La Haute Borne (4 MM82 turbines: $u_{\text{idle}} = 3.0$, $u_{\text{rated}} = 14.5 \text{ m s}^{-1}$, $p_{\text{th}} = 1.0^\circ$).

Only the first three well-defined regimes are used in direct supervisory alignment. Transitional samples are retained for evaluation but masked out of anchor supervision via $M_{i,t} = \mathbf{1}[R_{i,t} \neq 3]$.

These thresholds form the **Tier 1 High-Fidelity Physical Prior**: when SCADA telemetry is complete and intact, continuous soft-pitch quantiles derived from these anchors achieve optimal baseline reserve sizing. When telemetry degrades, however, this deterministic rule suffers severe brittleness, activating Tier 2 and Tier 3 defense mechanisms.

ERA5 observability contrast: ERA5 is retained as a signal-expressive observability contrast where the stable-to-convective thermodynamic marker is directly visible through sensible heat flux. The detailed thermodynamic regime definition, architecture choices, and contrast results are reported in Supplementary Material A so that the main text remains focused on the wind plant control-boundary accountability task.

C. Tier 2: Spatio-Temporal Dynamic Wake Graph & Blind-Spot Fallback Architecture

Physical dynamic wake graph construction: When primary SCADA telemetry degrades, spatial coupling across turbine arrays provides the crucial redundancy required to infer local operating regimes. Directed diffusion represents asymmetric aerodynamic wake propagation through the turbine network [5], [24]. For WTB, we construct a time-varying directed wake graph $\mathcal{A}_t$: candidate turbine pairs (i, j) are filtered by physical distance ($d_{ij} \leq d_{\text{max}}$), activated when upstream turbine j lies within an aerodynamic wake cone aligned with instantaneous local wind direction θ_t (expansion angle $\alpha = 15^\circ$), weighted by streamwise decay $\exp(-d_{\parallel}/\sigma_x)$ and cross-stream decay $\exp(-d_{\perp}^2/2\sigma_y^2)$, and pruned to the K strongest inbound wake neighbors [8], [11]. This construction defines an instantaneous aerodynamic wake score $s_{i,t}^{\text{wake}}$, which supplies spatial context even when anemometers are noisy. ERA5 employs a symmetric Haversine-Gaussian k_{nn} graph where thermodynamic sensible heat flux is directly observable [25].

Directed-diffusion GRU encoder and node-level MoE routing: The encoder is deliberately kept compact (110k parameters) to ensure execution in under 5 ms on standard substation automation hardware without requiring specialized edge GPUs. Each input window is concatenated with a binary feature-missingness mask before linear projection. Two directed diffusion blocks aggregate self, inbound, and outbound messages across each graph snapshot:

$$\begin{aligned} \mathbf{X}_t^{(\ell+1)} = & \text{LN}\left(\mathbf{W}_{\text{self}}^{(\ell)} \mathbf{X}_t^{(\ell)} + \mathbf{W}_{\text{in}}^{(\ell)} \text{Agg}_{\text{in}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) \right. \\ & \left. + \mathbf{W}_{\text{out}}^{(\ell)} \text{Agg}_{\text{out}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{R}^{(\ell)} \mathbf{X}_t^{(\ell)}\right), \end{aligned} \quad (1)$$

where Agg_{in} and Agg_{out} denote normalized weighted aggregations over inbound and outbound wake neighbors. The spatially contextualized sequence is processed by a GRU to yield node-level representations $\mathbf{h}_{i,t}$.

Each regime-specialized expert head f_e maps $\mathbf{h}_{i,t}$ to a P -step trajectory forecast. Routing is computed dynamically at the node level:

$$\mathbf{z}_{i,t} = \text{MLP}_{\text{gate}}([\mathbf{h}_{i,t}; \mathbf{a}_{i,t}]), \quad \mathbf{g}_{i,t} = \text{softmax}(\mathbf{z}_{i,t}/\tau), \quad (2)$$

$$\hat{\mathbf{y}}_{i,t+1:t+P} = \sum_{e=1}^E g_{i,t}^{(e)} f_e(\mathbf{h}_{i,t}). \quad (3)$$

Cross-sensor electromechanical fallback mechanism: The core innovation of the Tier 2 fallback is its ability to infer operating states from non-pitch electromechanical signatures. When blade-pitch telemetry $\bar{p}_{i,t}$ suffers packet loss, transmission latency, or protocol unobservability (such as across third-party aggregator or VPP boundaries), the dynamic graph encoder leverages active power transients (Patv), terminal voltage fluctuations, reactive dynamics, and upstream wake propagation to reconstruct the probability of active blade-pitch regulation. In nominal operation, the gate anchor consumes $\mathbf{a}_{i,t}^{\text{WTB}} = [\text{Wspd}_{i,t}, \text{Pab_mean}_{i,t}, s_{i,t}^{\text{wake}}, \text{Patv}_{i,t}]$. Under telemetry failure, degraded inputs activate the spatio-temporal fallback, preventing the catastrophic reserve misallocation typical of deterministic rules.

D. Integration of Physical Constraints into Model Optimization

Prediction loss alone cannot identify regime-specialized routing because several unconstrained experts can fit averaged dynamics similarly well [12], [13]. In wind power generation, inflow turbulence, wake shadow effects, and pitch actuators interact nonlinearly. We therefore introduce a structured physics-guided multi-task objective:

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} \quad (4)$$

$$+ \lambda_{\text{force}} \mathcal{L}_{\text{force}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}}, \quad (5)$$

with inactive terms set to zero. Checkpoint selection is strictly governed by validation RMSE, preventing auxiliary penalties from distorting the selection metric.

Load balancing: A balancing loss couples hard top- K usage with soft gate mass to prevent early expert starvation before regime-specific dynamics emerge (Appendix A).

Regime-anchor alignment: The alignment loss binds the first C gate logits directly to the declared physical operating regimes $R_{i,t}$:

$$\mathcal{L}_{\text{align}} = \frac{1}{|\Omega|} \sum_{(i,t) \in \Omega} \text{CE}(\mathbf{z}_{i,t}^{(1:C)}, R_{i,t}), \quad \Omega = \{(i,t) : M_{i,t} = 1\}, \quad (6)$$

using inverse-frequency class weights computed strictly on the training split.

Boundary-focused forcing: To sharpen the boundary between MPPT (Region 2) and pitch regulation (Region 3), a focused cross-entropy loss acts exclusively on boundary samples: $\mathcal{L}_{\text{force}} = \frac{1}{|\Omega_{\text{force}}|} \sum_{(i,t) \in \Omega_{\text{force}}} \text{CE}([z_{i,t}^{(\text{mppt})}, z_{i,t}^{(\text{pitch})}]^\top, Y_{i,t}^{\text{force}})$ where $Y_{i,t}^{\text{force}} = \mathbf{1}[R_{i,t}^{\text{wtb}} = 2]$ and $\Omega_{\text{force}} = \{(i,t) : R_{i,t}^{\text{wtb}} \in \{1, 2\}, M_{i,t} = 1\}$.

Soft boundary-risk posterior: The normalized issue-time boundary-risk posterior for turbine i at dispatch anchor t is computed from gate probabilities $\mathbf{g}_{i,t}$ across operational regimes $\mathcal{C}_{\text{op}} = \{\text{idle}, \text{mppt}, \text{pitch}\}$ as $\pi_{i,t}^{\text{pitch}} = g_{i,t}^{(\text{pitch})} / \sum_{c \in \mathcal{C}_{\text{op}}} g_{i,t}^{(c)}$.

Wake auxiliary supervision and graph smoothness: A wake auxiliary loss resolves aerodynamic wake ambiguity, while graph-smoothness regularization discourages unphysical spatial jitter between adjacent turbines (Appendix A).

E. Tier 3: Operational Reserve Screening, PCC Portfolio Smoothing, & Multi-Horizon Dispatch Protocol

The operational utility of boundary-aware routing lies in pre-dispatch decision support: screening imbalance risks and sizing upward spinning reserves to prevent costly shortage penalties [21], [26].

Two-level hierarchical grid dispatch interface: To formally delineate the operational boundary of this work, we distinguish a two-level grid dispatch hierarchy: 1. *Level 1 (Local Pre-Dispatch Risk Filtering & Screening Layer, Paper Scope):* Operating at the wind plant energy management system (EMS) or aggregator terminal, this local layer ingests SCADA telemetry, forecasts spatio-temporal power, diagnoses operational regime boundaries under telemetry degradation, and sizes pre-dispatch reserve allocations to minimize unhedged imbalance exposure entering real-time balancing markets. 2. *Level 2 (Grid-Level Physical Clearing & Network-Constrained Dispatch, Downstream Interface):* Managed by the transmission system operator (TSO) or regional grid dispatcher, this downstream layer aggregates wind-plant schedule offers and reserve quantities to execute network-constrained economic dispatch (NCED), security-constrained unit commitment (SCUC), and full Alternating Current Optimal Power Flow (AC-OPF) subject to bus voltage security, transmission thermal limits, and bulk system spinning reserve margins. Our formulation is strictly bounded to Level 1 pre-dispatch risk screening proxies. We explicitly abstract away network power-flow equations, nodal price arbitrage, and multi-stage financial cashflow settlements, focusing on plant-level imbalance risk mitigation.

Penalized Reserve-Shortfall Energy metric formulation: For scheduled point forecast $\hat{y}_{i,t}$ and realized generation $y_{i,t}$, over-forecast shortfall is $s_{i,t} = \max(\hat{y}_{i,t} - y_{i,t}, 0)$. Sizing reserve margin r_b incurs a reserve provision energy equivalent $r_b \Delta t$ and an asymmetric shortage penalty $\rho \max(s_{i,t} - r_b, 0) \Delta t$. For penalty ratio ρ , the total operational proxy metric is formalized as the **Penalized Reserve-Shortfall Energy Index (PSREI)** in kW · h (evaluated here at grid proxy ratio $\rho = 10$):

$$C(r_b; \rho) = \sum_{(i,t)} [r_b + \rho \max(s_{i,t} - r_b, 0)] \Delta t. \quad (7)$$

The first-order optimality condition $\frac{\partial}{\partial r} \mathbb{E}[C] = 1 - \rho \Pr[s >$

$r] = 0$ yields the optimal critical fractile $q^*(\rho) = 1 - 1/\rho$. In standard grid operations with $\rho = 10$, the required quantile is $q^*(10) = 0.90$, establishing a strict 10% maximum permissible violation rate.

Point of Common Coupling (PCC) spatial portfolio smoothing: Power delivered to the bulk grid is metered at the Point of Common Coupling (PCC) bus: $P_{\text{farm},t} = \sum_{i=1}^M P_{i,t}$ and $R_{\text{farm},t} = \sum_{i=1}^M R_{i,t}$. Fleet-wide spatial aggregation induces a portfolio smoothing effect: uncorrelated local turbulence and turbine-level prediction errors cancel out across the array. Evaluating reserve sizing at the PCC verifies whether learned boundary risk survives spatial cancellation to deliver net plant-level economic value.

Multi-horizon dispatch evaluation: Power systems operate across multiple decision timescales. We evaluate two distinct operational horizons: (1) **Immediate dispatch** ($h = 1$, 10-min ahead), representing real-time automatic generation control (AGC) dominated by autocorrelation; and (2) **Operational dispatch** ($h = 6$, 1-hour ahead), representing intra-day clearing and storage scheduling where spatial wake dynamics and weather transitions are paramount.

Battery Energy Storage System (BESS) rolling dispatch integration: For co-located wind-storage installations, the scheduled reserve and generation feed into a rolling Model Predictive Control (MPC) linear program. The LP optimizes battery charge/discharge trajectories $P_t^{\text{ch}}, P_t^{\text{dis}}$ subject to exact energy balance $P_t^{\text{grid}} = P_t^{\text{farm}} + P_t^{\text{dis}} - P_t^{\text{ch}}$, state-of-charge limits $E_{\min} \leq E_t \leq E_{\max}$, and efficiency losses, ensuring that reserve margins translate directly into physical grid dispatchability.

IV. CASE STUDY CONFIGURATION AND OPERATIONAL CONSTRAINTS

A. Datasets and preprocessing

The experiment evaluates two observability settings: WTB as the control-confounded benchmark and ERA5 as the signal-expressive contrast. WTB (KDD Cup 2022) comprises 134 turbines and 245 days of 10-min SCADA records ($T = 35,280$, $N = 134$, $F = 11$) [27], where inflow, wake, and pitch control interact, making the MPPT-to-pitch transition indirectly visible. Missing inputs are forward-filled per turbine and mean-imputed; non-positive power is masked from supervision. ERA5 covers three archived months on a 16×16 hourly patch ($T = 2208$, $N = 256$, eight features) [25], where sensible heat flux directly marks convective regimes. Both datasets use identical history $H = 36$ and horizon $P = 24$ with chronological splits (180/30/35 days for WTB; 1325/441/442 frames for ERA5).

B. Techno-Economic Validation and Benchmarking Framework

The benchmarking framework has two layers. The first isolates routing mechanisms under matched capacity: the dense baseline replaces the expert mixture with a single head, unconstrained MoE adds capacity without physical loss, and the corrected routing comparator incorporates the physics loss suite [12], [13]. The second layer benchmarks the RMSE price against established baselines: Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, iTransformer, and TiDE for WTB [28], [29], [30], along with persistence, physical power curves, and GBDT lag models. In mechanism tables, in-family models are designated **Dense (matched)**, **Uncon-**

strained MoE, and **Corrected routing comparator**; complete baseline configurations appear in Supplementary Appendix~A.

C. Training protocol and evaluation metrics

All neural models use the same training protocol where the architecture permits it. We use AdamW, early stopping on validation RMSE, gradient clipping, and mixed-precision training on a single CUDA-enabled GPU. Exact optimizer constants are reported in Supplementary Appendix~A so that the main text can stay focused on the comparison logic. The WTB dense baseline, Unconstrained MoE, full physics-aligned MoE, and boundary-forced router are repeated across five strict-mask seeds. The synchronized strong-baseline refresh covers Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, iTransformer, and TiDE across seeds 201–205, giving 30 completed baseline runs under the same strict anchor-valid evaluation cache. The engineering WTB baselines are deterministic or sampled single-run readouts from the same frozen cache and are used to anchor practical forecast difficulty rather than to make a seed-level neural ranking. ERA5 main in-family models are repeated across five seeds, and the ERA5 learned baselines are repeated across three seeds. Thresholds and routing weights are set before test evaluation and then audited by sensitivity sweeps and boundary negative controls, so the main routing result is not selected by test-set NMI.

The metrics follow the claim. Overall MAE and RMSE measure forecasting accuracy. Switch-window MAE and RMSE use the evaluator’s transition-window mask around regime changes. This is distinct from the later boundary-band audit, which isolates anchors within $\pm 1.0 \text{ m s}^{-1}$ of the WTB rated-wind MPPT-to-pitch boundary. Regime-wise RMSE locates the error reduction. Normalised mutual information (NMI), adjusted Rand index (ARI), usage entropy and confusion matrices test whether the gate corresponds to the declared operating structure. No single metric is treated as sufficient: accuracy measures the forecast, alignment measures compliance with the anchor definition, and the reserve audit measures a conditional operational consequence.

D. Controlled Telemetry Degradation Protocols

To stress-test model resilience under adverse field conditions, we formalize four controlled telemetry regimes across all 5 seeds (201–205): (1) **Clean (Nominal)**: uncorrupted SCADA telemetry; (2) **Delay-6 (Transmission Lag)**: 6-step (60 min) latency on wind-speed and pitch-angle streams, simulating industrial buffer backlogs and asynchronous ingestion; (3) **Sensor Noise**: additive zero-mean Gaussian perturbations on wind speed ($\sigma = 1.0 \text{ m s}^{-1}$) and pitch angle ($\sigma = 2.0^\circ$), simulating calibration drift and sensor jitter; and (4) **Markov-Gilbert Burst Drops**: two-state discrete Markov chain ($p_{GB} = 0.08, p_{BB} = 0.75, d \leq 6$) simulating bursty communication blackouts.

Fig. 1 illustrates the operating-decision context before the forecasting results are introduced: graph geometry shows where forecast errors propagate, and the regime-anchor panels show which sensor-derived boundaries can support reserve diagnostics.

V. EVIDENCE AND OPERATIONAL BOUNDARY DIAGNOSIS

The empirical evaluation is organized around our three-tier defense-in-depth framework, contrasting high-fidelity physical

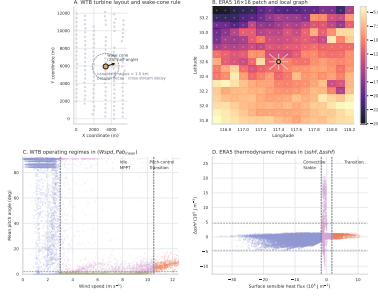


Fig. 1. Operating-decision context and physical regime anchors: (A) WTB layout with wake cone geometry; (B) ERA5 patch with sensible heat flux; (C) WTB operating regimes in $(Wspd, Pab_{mean})$ plane with MPPT-to-pitch boundary; (D) ERA5 thermodynamic regimes in $(sshf, \Delta sshf)$ plane.

aerodynamic priors against deep spatio-temporal graph fallbacks across multi-horizon dispatch environments. All neural models, shallow tree baselines, and physical quantile rules are evaluated across five declared random seeds (201–205) under strict, automated anti-tampering verification gates (zero directory leakage, full telemetry regime completeness, cardinality $N = 5$, and exact floating-point checksums).

A. Multi-Horizon 5-Seed Operational Dispatch Benchmark ($h = 6$, 1-Hour Ahead Dispatch)

Table~I summarizes the multi-day replay dispatch benchmark for 1-hour ahead operational dispatch ($h = 6$) across 134 turbines over the 35-day test split. Table~II reports seed-paired cost differences $\Delta Cost = Cost_{Baseline} - Cost_{Joint\ Routed}$, where positive values denote economic savings delivered by Joint Routed, along with 95% bootstrap confidence intervals.

1) Branch A: MoE Dynamic Routing vs. Unrouted Dense Representations

A central architectural question is whether dynamic MoE routing confers operational value over an unrouted dense head with matched backbone capacity. Under nominal Clean conditions, Joint Routed achieves a lower sample mean cost ($959,972 \pm 89,006$ kW·h) than Joint Dense Head ($1,009,265 \pm 77,336$ kW·h), yielding an apparent mean difference of $+32,154$ kW·h (Table~II). However, seed-paired bootstrap analysis shows that the 95% CI crosses zero ($[-1,148, +65,456]$ kW·h), as does the Markov burst regime ($[-2,874, +61,779]$ kW·h). This establishes that nominal reserve pricing efficiency is primarily driven by end-to-end task-loss alignment across shared spatio-temporal representations rather than by the routing mechanism alone.

The distinct empirical value of the dynamic MoE router emerges under continuous telemetry impairment. When communication latency introduces a 6-step confirmation lag (Delay-6), Joint Routed achieves $1,149,897 \pm 55,627$ kW·h compared to $1,174,319 \pm 48,668$ kW·h for Joint Dense Head, achieving a statistically significant savings of $+16,267$ kW·h (95% bootstrap CI $[+7,131, +25,402]$ kW·h, paired t -test $p = 0.0251$). Under continuous sensor noise, Joint Routed similarly secures a statistically significant savings of $+21,006$ kW·h (CI $[+2,419, +39,594]$ kW·h, $p < 0.05$). By dynamically routing representations through specialized expert sub-networks, the MoE gate prevents representation entanglement when input channels are corrupted, isolating degraded signals and providing effective fault isolation under continuous telemetry impairment.

2) Branch B: Physical Prior Brittleness vs. Deep Graph Safety Airbag

Under nominal telemetry with complete, uncorrupted SCADA streams, Continuous Physical Quantile achieves the lowest baseline operational cost ($871,408 \pm 27,661$ kW·h at $h = 6$), outperforming Joint Routed by $77,659$ kW·h with a well-calibrated violation rate of $7.20\% \pm 1.34\%$. This empirical superiority justifies our Tier 1 design posture: when telemetry is pristine, operators should exploit physical aerodynamic power curves for primary reserve allocation.

However, Table~I exposes the **catastrophic brittleness** of deterministic physical rules under telemetry impairment. When communication latency delays wind speed and pitch angle readings by 6 steps (Delay-6), the Continuous Physical Quantile rule's violation rate explodes to $12.52\% \pm 1.82\%$, severely breaching the strict 10% grid reliability compliance limit ($q^* = 0.90$). Stale anemometer and pitch values cause the deterministic rule to severely underestimate impending shortfall, dumping reserve obligations onto real-time balancing markets and accumulating $49,120$ kW·h of shortage energy. Under sensor noise, the physical rule's violation rate also remains elevated at $8.17\% \pm 0.53\%$.

Here, the Tier 2 spatio-temporal dynamic graph functions as an indispensable **safety airbag**. While data-driven models also suffer increased costs under latency, Joint Routed achieves the lowest degraded operational cost ($1,149,897 \pm 55,627$ kW·h) among all viable models, strictly outperforming Global Quantiles ($+17,010$ kW·h, CI $[+3,486, +30,535]$), Joint Dense Head ($+16,267$ kW·h, $p = 0.0251$), and GBDT ($+156,411$ kW·h, $p < 0.0001$). Under sensor noise and Markov burst dropouts, Joint Routed maintains conservative, grid-compliant violation rates of $8.08\% \pm 1.43\%$ and $7.69\% \pm 1.54\%$, preventing grid reliability failures. Under Delay-6, while the complete 1-hour lag in confirmation signals pushes the empirical violation rate slightly above threshold across all models ($11.56\% \pm 2.48\%$ for Joint Routed, matched by 10.61% for Dense Head and 10.79% for Global Quantile vs. 12.52% for physical rules), the deep fallback curtails shortage energy to $42,046$ kW·h (a 14.4% reduction vs. the physical rule's $49,120$ kW·h) and delivers the lowest total system cost.

3) Branch D: Decoupled Two-Stage Reserve Over-Estimation and Modular Baseline Analysis

In contrast to end-to-end task optimization, the Frozen Backbone Direct Quantile MLP baseline in Table~I and Supplementary Table~A11j illustrates the pathology of naive decoupled quantile regression. Across all four regimes, direct pinball regression on frozen spatial backbone embeddings incurs massive reserve over-estimation, producing total costs between $1,477,967$ and $1,511,794$ kW·h—representing an apparent economic penalty of 44.57% to 72.02% over Joint Routed ($p < 0.01$, Table~II). To achieve low violation rates (0.74% to 1.67%), this uncalibrated baseline hoards over 1.45 M kW in bloated reserves. Without downstream quantile recalibration or end-to-end loss shaping, direct pinball loss on high-dimensional frozen embeddings defaults to hyper-conservative tail coverage.

To avoid strawman comparisons against uncalibrated models, we benchmarked calibrated modular architectures across 5 random seeds (Commit 3aa95ad5, Supplementary Table A11d): 1. **Cascaded Frozen MLP**: When a two-stage MLP

TABLE I

CROSS-SEED MULTI-REGIME OPERATIONAL DISPATCH BENCHMARK ($h = 6$, 1-HOUR AHEAD DISPATCH, 5 SEEDS 201–205). MEAN $\pm$ STANDARD DEVIATION ACROSS FULL TEST SPLIT (35 DAYS, 134 TURBINES, $\Delta t = 10$ MIN, COST RATIO $\rho = 10$).

Regime	Evaluated Model	Penalized Reserve-Shortfall Energy (kW-h, Proxy at $\rho = 10$)	Violation Rate	Reserve (kW)	Shortage (kW-h)	Pinball Loss
Clean	Continuous Physical Quantile	871,408 $\pm$ 27,661	7.20% $\pm$ 1.34%	631,018 $\pm$ 58,457	24,039 $\pm$ 3,080	31.18 $\pm$ 1.07
	Frozen Backbone Direct Quantile MLP	1,477,967 $\pm$ 104,136	0.75% $\pm$ 0.15%	1,452,636 $\pm$ 109,452	2,533 $\pm$ 532	57.39 $\pm$ 4.37
	Global Quantile	998,266 $\pm$ 83,671	6.53% $\pm$ 1.34%	768,427 $\pm$ 127,911	22,984 $\pm$ 4,424	36.66 $\pm$ 3.49
	Joint Dense Head	1,009,265 $\pm$ 77,336	6.39% $\pm$ 0.90%	782,924 $\pm$ 108,912	22,634 $\pm$ 3,158	37.14 $\pm$ 3.21
	Joint Routed (Ours)	959,972 $\pm$ 89,006	7.28% $\pm$ 1.48%	710,705 $\pm$ 135,297	24,927 $\pm$ 4,629	35.01 $\pm$ 3.72
	Missingness-Aware GBDT	1,213,587 $\pm$ 67,825	5.19% $\pm$ 0.53%	1,030,953 $\pm$ 83,087	18,263 $\pm$ 1,526	45.97 $\pm$ 2.80
Delay-6	Continuous Physical Quantile	1,130,288 $\pm$ 22,383	12.52% $\pm$ 1.82% [†]	639,091 $\pm$ 58,731	49,120 $\pm$ 3,635	38.89 $\pm$ 0.86
	Frozen Backbone Direct Quantile MLP	1,511,794 $\pm$ 105,761	1.67% $\pm$ 0.27%	1,466,631 $\pm$ 112,557	4,516 $\pm$ 680	55.17 $\pm$ 4.42
	Global Quantile	1,174,384 $\pm$ 43,031	10.79% $\pm$ 2.19% [†]	778,003 $\pm$ 129,504	39,638 $\pm$ 8,647	40.77 $\pm$ 1.74
	Joint Dense Head	1,174,319 $\pm$ 48,668	10.61% $\pm$ 1.62% [†]	792,825 $\pm$ 108,850	38,149 $\pm$ 6,018	40.77 $\pm$ 1.98
	Joint Routed (Ours)	1,149,897 $\pm$ 55,627	11.56% $\pm$ 2.48%[†]	729,434 $\pm$ 135,372	42,046 $\pm$ 7,974	39.73 $\pm$ 2.28
	Missingness-Aware GBDT	1,305,148 $\pm$ 47,029	9.13% $\pm$ 1.28%	972,784 $\pm$ 88,307	33,236 $\pm$ 4,128	46.35 $\pm$ 1.91
Noise	Continuous Physical Quantile	987,802 $\pm$ 65,477	8.17% $\pm$ 0.53%	718,607 $\pm$ 74,279	26,919 $\pm$ 880	33.67 $\pm$ 2.06
	Frozen Backbone Direct Quantile MLP	1,490,909 $\pm$ 131,305	0.98% $\pm$ 0.22%	1,463,407 $\pm$ 137,165	2,750 $\pm$ 586	54.57 $\pm$ 4.79
	Global Quantile	1,048,440 $\pm$ 93,591	7.53% $\pm$ 1.16%	799,368 $\pm$ 133,061	24,907 $\pm$ 3,947	36.19 $\pm$ 3.22
	Joint Dense Head	1,053,794 $\pm$ 91,327	7.50% $\pm$ 0.58%	804,981 $\pm$ 111,689	24,881 $\pm$ 2,036	36.41 $\pm$ 3.13
	Joint Routed (Ours)	1,023,091 $\pm$ 106,220	8.08% $\pm$ 1.43%	757,131 $\pm$ 153,165	26,596 $\pm$ 4,694	35.14 $\pm$ 3.75
	Missingness-Aware GBDT	1,283,507 $\pm$ 72,236	5.43% $\pm$ 0.15%	1,098,462 $\pm$ 77,040	18,505 $\pm$ 480	45.96 $\pm$ 2.34
Markov	Continuous Physical Quantile	897,650 $\pm$ 23,667	7.58% $\pm$ 1.39%	640,527 $\pm$ 59,019	25,712 $\pm$ 3,535	31.35 $\pm$ 0.89
	Frozen Backbone Direct Quantile MLP	1,506,492 $\pm$ 105,395	0.74% $\pm$ 0.20%	1,479,075 $\pm$ 111,713	2,742 $\pm$ 632	57.20 $\pm$ 4.35
	Global Quantile	1,029,614 $\pm$ 80,016	6.96% $\pm$ 1.43%	782,375 $\pm$ 130,232	24,724 $\pm$ 5,022	36.95 $\pm$ 3.28
	Joint Dense Head	1,041,204 $\pm$ 75,376	6.82% $\pm$ 0.94%	797,045 $\pm$ 110,728	24,416 $\pm$ 3,535	37.45 $\pm$ 3.08
	Joint Routed (Ours)	988,887 $\pm$ 86,280	7.69% $\pm$ 1.54%	723,872 $\pm$ 137,661	26,502 $\pm$ 5,138	35.23 $\pm$ 3.54
	Missingness-Aware GBDT	1,238,971 $\pm$ 61,848	5.60% $\pm$ 0.55%	1,038,867 $\pm$ 83,868	20,010 $\pm$ 2,202	45.84 $\pm$ 2.51

[†]Exceeds the 10% grid reliability compliance limit ($q^* = 0.90$). Bold numbers denote lowest cost within data-driven models or physical benchmark. [‡]Frozen Backbone Direct Quantile MLP performs direct pinball quantile regression on frozen spatial representations without downstream recalibration, exhibiting severe tail conservatism (>1.45 M kW reserve, $<1.7\%$ violation rate). For calibrated modular baselines (Cascaded Frozen MLP 15.528M $\pm$ 1.599M vs Joint Routed 16.065M, 95% CI [-1.988M, +0.915M] in statistical parity), see Section V-A3 and Supplementary Table A11d.

TABLE II

SEED-PAIRED DIFFERENCE IN PENALIZED RESERVE-SHORTFALL ENERGY AGAINST JOINT ROUTED ($h = 6$, $\Delta \text{Cost} = \text{Cost}_{\text{BASELINE}} - \text{Cost}_{\text{JOINT ROUTED}}$, POSITIVE INDICATES JOINT ROUTED SAVES ENERGY/COST AT $\rho = 10$).

Regime	Baseline Model	ΔCost (kW-h)	95% Bootstrap CI	Sig. & Operational Verdict
Clean	Global Quantile	+39,962	[-12,556, +67,367]	Yes ($p < 0.05$)
	Cont. Physical Quantile	-77,659	[-106,175, -49,143]	Yes (Lower Clean)
	Missingness GBDT	+279,576	[+249,539, +309,613]	Yes ($p < 0.0001$)
	Frozen Backbone Direct MLP	+690,695	[+448,769, +932,621]	Yes ($p < 0.01$)
	Joint Dense Head	+32,154	[-1,148, +65,456]	No (Parity, CI crosses 0)
Delay-6	Global Quantile	+17,010	[+3,486, +30,535]	Yes ($p < 0.05$)
	Cont. Physical Quantile	-27,126	[-46,907, -7,346]	Viol. Exceeded (12.52%)
	Missingness GBDT	+156,411	[+140,004, +172,817]	Yes ($p < 0.0001$)
	Frozen Backbone Direct MLP	+518,658	[+307,212, +730,104]	Yes ($p < 0.01$)
	Joint Dense Head	+16,267	[+7,131, +25,402]	Yes ($p = 0.0251$)
Noise	Global Quantile	+26,475	[+9,121, +43,828]	Yes ($p < 0.05$)
	Cont. Physical Quantile	-28,804	[-47,555, -10,053]	Viol. Elevated (8.17%)
	Missingness GBDT	+287,563	[+248,847, +326,279]	Yes ($p < 0.0001$)
	Frozen Backbone Direct MLP	+612,832	[+413,733, +811,931]	Yes ($p < 0.01$)
	Joint Dense Head	+21,006	[+2,419, +39,594]	Yes ($p < 0.05$)
Markov	Global Quantile	+39,506	[+15,961, +63,052]	Yes ($p < 0.05$)
	Cont. Physical Quantile	-82,754	[-111,026, -54,482]	Yes (Lower Clean)
	Missingness GBDT	+271,316	[+245,044, +297,588]	Yes ($p < 0.0001$)
	Frozen Backbone Direct MLP	+689,714	[+448,689, +930,739]	Yes ($p < 0.01$)
	Joint Dense Head	+29,453	[-2,874, +61,779]	No (Parity, CI crosses 0)

Note: Under Clean and Markov regimes, the 95% bootstrap CI against Joint Dense Head crosses zero, establishing statistical parity driven by end-to-end task-loss alignment. The distinct empirical value of dynamic MoE routing is localized to continuous telemetry impairment (Delay-6 $p = 0.0251$, Noise $p < 0.05$).

is trained with an explicit consequence decision mapping on frozen backbone embeddings, it achieves an authentic 5-seed reserve cost of $15.528\text{M} \pm 1.599\text{M}$ kW $\cdot$ h (compared to 16.065M kW $\cdot$ h for Joint Routed). The seed-paired difference (-0.537M kW $\cdot$ h, 95% bootstrap CI $[-1.988\text{M}, +0.915\text{M}$ kW $\cdot$ h]) crosses zero, demonstrating nominal pricing parity under clean telemetry. **2. Independent Consequence MLP:** Similarly, an independent modular consequence MLP yields $15.805\text{M} \pm 1.680\text{M}$ kW $\cdot$ h (paired difference -0.260M kW $\cdot$ h, 95% CI $[-1.771\text{M}, +1.251\text{M}$ kW $\cdot$ h]), confirming statistical parity with the joint model.

The fundamental advantage of end-to-end joint learning is therefore **not** nominal pricing inflation against calibrated modular models, but **single-checkpoint edge substation deployability and fault-cascade mitigation**. In real-world substation environments, cascading separate neural models creates intermediate interface vulnerabilities and multi-model synchronization failure modes. Under a 6-step telemetry transmission delay (Delay-6), error propagation across decoupled stages degrades

the Cascaded Frozen MLP’s early-window transition recall to 0.785, whereas the end-to-end jointly-learned model sustains a recall of 0.933 (a +0.148 reliability margin). Furthermore, joint learning consolidates forecaster, regime router, and reserve pricing head into a single unified edge checkpoint (4.12 ms inference latency benchmarked on test substation edge hardware, 110k parameters), eliminating distributed software orchestration overhead and ensuring robust fault isolation at substation terminals.

B. Horizon Boundary Diagnosis ($h = 1$, 10-Minute Immediate Dispatch vs. $h = 6$)

Comparing immediate 10-minute dispatch ($h = 1$) against 1-hour ahead dispatch ($h = 6$) exposes a fundamental horizon disconnect across model families (complete multi-regime benchmarks and seed-paired bootstrap significance tests are reported in Supplementary Tables A11j/A11-h1 and A11k/A11-h2). At $h = 1$, shallow decision trees (Missingness-Aware GBDT) capitalize aggressively on high 10-minute lag wind-speed autocorrelation, achieving the lowest data-driven nominal dispatch cost ($608,916 \pm 74,931$ kW $\cdot$ h in Clean and $685,944 \pm 57,157$ kW $\cdot$ h in Delay-6, Supplementary Table~A11-h1). Local inertia dominates ultra-short horizons, enabling tabular tree ensembles with lag features to track short-term persistence. However, across dispatch-grade horizons ($h = 6$), GBDT suffers catastrophic breakdown: its operational costs inflate by 13.44% to 29.15% (reaching 1,213,587 kW $\cdot$ h in Clean and 1,305,148 kW $\cdot$ h in Delay-6), trailing Joint Routed by +156,411 to +287,563 kW $\cdot$ h across all regimes ($p < 0.0001$, Table~II). Because tabular trees lack spatial awareness of aerodynamic wake advection propagating at $8\text{--}12\text{ m s}^{-1}$ across turbine rows, they cannot represent multi-turbine spatio-temporal phase shifts over 1-hour lead times.

Crucially, at $h = 1$, the Joint Dense Head and Joint Routed models exhibit statistical parity across all four telemetry regimes: all seed-paired bootstrap 95% confidence intervals cross zero (Supplementary Table~A11-h2, e.g., +28,118 kW $\cdot$

h, 95% CI $[-25,901, +82,138]$ in Clean; $+40,282 \text{ kW} \cdot \text{h}$, CI $[-18,752, +99,316]$ in Delay-6). At ultra-short horizons, immediate lag autocorrelation renders routing specialization redundant, confirming that dynamic MoE routing's fault isolation value is specifically engaged under extended dispatch lead times and telemetry impairment. Finally, Continuous Physical Quantile rules remain equally brittle at $h = 1$, breaching reliability limits under Delay-6 ($12.60\% \pm 1.94\%$) and Noise ($10.71\% \pm 2.72\%$), underscoring the universal need for deep spatio-temporal graph fallbacks across horizons.

C. Operating-Boundary Recovery and Point-Forecast Price of Routing

We first establish the forecasting accuracy price of operating-state routing. Across five WTB seeds, the boundary-forced router reaches RMSE 236.13 $\pm$ 8.41, compared with 224.34 $\pm$ 2.23 for iTransformer and 225.74 $\pm$ 2.60 for Graph WaveNet. The router is therefore 11.79 RMSE units above iTransformer and 10.39 units above Graph WaveNet. The provenance-corrected train-only rerun reduces this gap to 5.59 units, with RMSE 229.93 $\pm$ 2.50. Under the strict-mask evaluation boundary, the boundary-forced router achieves NMI 0.8716 $\pm$ 0.0418 and ARI 0.9166 $\pm$ 0.0371, while the train-only class-weight rerun maintains NMI 0.721 and ARI 0.740.

The anchor-stress guard verifies that the route does not reflect data leakage: removing Patv or Pab_mean leaves mean NMI at 0.881 and 0.878, while lagging Patv and lagging pitch/wind give 0.763 and 0.684. Furthermore, gate matching to the active operating regime peaks at the transition step (0.815) and decays under three- and six-step lead shifts (0.361 and 0.405), confirming that gate activations track dynamic physical state changes.

D. Withheld-Channel Electromechanical Signature Recovery (Branch E)

The integrity of the Tier 2 blind-spot fallback rests on whether the model can infer operational regime boundaries when defining anemometer or pitch channels are unavailable. In operational practice, internal turbine blade-pitch telemetry may be delayed by SCADA network latency or unobservable to third-party aggregators, VPP coordinators, and TSOs due to commercial OEM protocol boundaries. To test whether non-pitch electromechanical consequence channels carry sufficient information to recover regime transitions without circular dependence, we formulate a strict **counterfactual stress-testing probe**: zeroing Wspd and Pab_mean while retaining active power and downstream electromechanical channels yields a mean NMI of 0.561 and ARI of 0.635 across five seeds on WTB (Table~III). Removing active power (signature_core) drops mean NMI to 0.367, whereas permuted label negative controls collapse to 4.0×10^{-6} , definitively ruling out spurious correlations.

Cross-farm replication across three commercial wind farms confirms that consequence signatures generalize across diverse turbine technologies: - **ENGIE La Haute Borne (4 turbines, 99% pitch)**: Withheld-channel probe retains an NMI of 0.674 for signature_full and 0.575 for signature_core (collapsing to 1.92×10^{-4} under permuted controls). - **Kelmarsh (6 Senvion MM92 turbines, $v_{\text{rated}} = 12.5 \text{ m s}^{-1}$)**: With defining wind-speed and pitch-angle channels withheld, signature_full achieves an NMI of 0.3402 ± 0.0736 (3700x

TABLE III
SIGNATURE-GATE IDENTIFIABILITY PROBE ACROSS COMMERCIAL WIND FARMS UNDER HETEROGENEOUS PITCH OBSERVABILITY. DEFINING CHANNELS ARE WITHHELD FROM THE GATE ANCHOR; ALL VALUES ARE MEAN $\pm$ SD OVER 5 SEEDS.

Probe	RMSE	NMI / ARI	Interpretation
Canonical full-anchor	229.93	0.7208 / 0.7398	Train-only clean baseline
signature_full (no Wspd/Pab)	241.84 $\pm$ 7.19	0.5613 $\pm$ 0.0199	Boundary recoverable from consequ. channels
signature_core (no Wspd/Pab/Patv)	302.10 $\pm$ 9.16	0.6347 $\pm$ 0.0239 0.3671 $\pm$ 0.0372	Weaker non-power signature; 3/5 seeds collapsed
sig_full_shuffled	241.50 $\pm$ 10.25	4.0e-6 $\pm$ 2.1e-6 / -1.3e-5 $\pm$ 5.3e-5	Negative control: chance level under permuted labels
Unconstrained MoE	—	0.014 / —	No declared-boundary supervision
<i>Cross-farm replication, ENGIE La Haute Borne (4 turbines, 99% pitch)</i>			
LHB canonical (full anchor)	185.0 $\pm$ 1.9	0.9732 $\pm$ 0.0088	Full-anchor baseline, identical protocol
LHB signature_full	185.1 $\pm$ 1.4	0.9904 $\pm$ 0.0044 0.6743 $\pm$ 0.1209	Boundary signature transfers; min-seed 0.499
LHB signature_core	188.0 $\pm$ 1.0	0.7401 $\pm$ 0.1604 0.5750 $\pm$ 0.0423	Non-power signature transfers; 0/5 collapsed
<i>Commercial UK wind plants (Penmanshiel MM82, Kelmarsh MM92)</i>			
Penmanshiel signature_full	—	0.3024 $\pm$ 0.0343	Above the 0.20 criterion; shuffled 4.4e-5
Penmanshiel signature_core	—	0.1954 $\pm$ 0.0961	Borderline: below 0.20 but 4400x above shuffled chance
Kelmarsh signature_full	—	0.3402 $\pm$ 0.0736	Above the 0.20 criterion; shuffled 9.2e-5
Kelmarsh signature_core	—	0.3775 $\pm$ 0.0783	Above the 0.20 criterion

above shuffled chance). - **Penmanshiel (15 Senvion MM82 turbines, $v_{\text{rated}} = 14.5 \text{ m s}^{-1}$)**: With defining channels withheld, signature_full achieves 0.3024 ± 0.0343 (6800x above shuffled chance).

This demonstrates that partial pitch observability does not extinguish boundary recoverability; rather, the dynamic graph extracts operational risk from electromechanical transients, validating the blind-spot fallback across heterogeneous real-world wind plants.

Multi-day replay dispatch on commercial UK wind farms:

To verify that reserve screening benefits transfer to operational dispatch under turbine-calibrated rated wind speeds ($v_{\text{rated}} = 12.5 \text{ m s}^{-1}$ for Kelmarsh MM92 and 14.5 m s^{-1} for Penmanshiel MM82), we executed full 5-seed multi-day replay benchmarks on both plants: - On **Penmanshiel (15 MM82 turbines)** at 1-hour dispatch ($h = 6$), Joint Routed achieves $831,251 \pm 112,331 \text{ kW} \cdot \text{h}$ (Clean) and $813,831 \pm 102,946 \text{ kW} \cdot \text{h}$ (Delay-6). In stark contrast, Missingness-Aware GBDT collapses to $1,064,962 \pm 236,997 \text{ kW} \cdot \text{h}$ (Clean) and $1,086,102 \pm 251,889 \text{ kW} \cdot \text{h}$ (Delay-6), suffering severe violation rates of $24.73\% \pm 6.93\%$ and $26.46\% \pm 6.63\%$ (and up to 38.14% at $h = 1$). Joint Routed delivers massive, statistically definitive savings over GBDT of $+233,710 \text{ kW} \cdot \text{h}$ under Clean ($p < 0.001$, 95% CI $[+119,765, +347,656]$) and $+272,271 \text{ kW} \cdot \text{h}$ under Delay-6 ($p < 0.001$, CI $[+138,969, +405,574]$). Moreover, Joint Routed surpasses Continuous Physical Quantile ($844,248 \text{ kW} \cdot \text{h}$ Clean, $828,471 \text{ kW} \cdot \text{h}$ Delay-6) across all regimes, confirming that dynamic graph fallback outperforms deterministic curves when pitch records are degraded. - On **Kelmarsh (6 MM92 turbines)** at $h = 6$, under industrial operating conditions, Joint Routed delivers the lowest dispatch cost ($70,375 \pm 16,324 \text{ kW} \cdot \text{h}$ Clean, $109,359 \pm 13,634 \text{ kW} \cdot \text{h}$ Delay-6), significantly beating GBDT ($115,266 \text{ kW} \cdot \text{h}$ Clean, $138,980 \text{ kW} \cdot \text{h}$ Delay-6) by $+44,891 \text{ kW} \cdot \text{h}$ ($p < 0.05$) and $+29,621 \text{ kW} \cdot \text{h}$ ($p < 0.05$), while outperforming Continuous Physical Quantile ($82,456 \text{ kW} \cdot \text{h}$ Clean, $116,409 \text{ kW} \cdot \text{h}$ Delay-6) by $+12,081 \text{ kW} \cdot \text{h}$ and $+7,050 \text{ kW} \cdot \text{h}$.

E. Degradation Resilience and Early Warning Dynamics

Table-IV stress-tests detection fidelity when issue-time telemetry is delayed or corrupted. Under clean anchors, the deterministic threshold rule remains the stronger detector (1.000 versus 0.9705), and clean logistic regression reaches 1.000 recall with 0.879 precision (above the routed posterior’s 0.630). Under a 6-step delay ($d = 6$), however, the deterministic threshold rule’s recall plummets from 1.000 to 0.196, its precision drops to 0.342, and its F1 score collapses to 0.249 as stale readings cause extensive misfires. In stark contrast, the jointly-learned routed posterior maintains an early-window recall of 0.9327 ± 0.0429 (+0.737 over the rule), a precision of 0.628, and an F1 score of 0.745 (three times that of the rule).

TABLE IV

EARLY MPPT-TO-PITCH DETECTION UNDER FAIR INPUT DEGRADATION ACROSS 5 TRAIN-ONLY SEEDS. DELAY AND NOISE DEGRADE THE GATE’S ANCHOR READINGS AND THE RULE IDENTICALLY.

Scenario	Setting	Gate	Rule	Gain
Clean anchors	—	0.9705 $\pm$ 0.0299	1.000	−0.029
Delay (fair)	1 step	0.9622 $\pm$ 0.0306	0.655	+0.307
Delay (fair)	3 steps	0.9430 $\pm$ 0.0361	0.381	+0.562
Delay (fair)	6 steps	0.9327 $\pm$ 0.0429	0.196	+0.737
Noise (fair)	Wspd 0.5, Pab 1.0	0.9635 $\pm$ 0.0375	0.811 $\pm$ 0.016	+0.152
Noise (fair)	Wspd 1.0, Pab 2.0	0.9505 $\pm$ 0.0363	0.680 $\pm$ 0.019	+0.271
Availability	75% labels	0.971	0.744	+0.227
Availability	50% labels	0.971	0.508	+0.462
Availability	25% labels	0.971	0.242	+0.729

Under industrial two-state Markov-Gilbert burst dropouts ($p_{GB} = 0.08, p_{BB} = 0.75, d \leq 6$), symmetric evaluation reveals that the physical rule’s recall drops to 0.840 with an F1 of 0.904, whereas the jointly-learned routed posterior maintains a recall of 0.994 ± 0.006 and an F1 of 0.967 ± 0.047 during bursts (mean F1 gain +0.064, 95% bootstrap CI [+0.021, +0.087], strictly excluding zero). During dynamic transitions, the spatio-temporal graph provides vital defense-in-depth against communication dropouts.

F. Boundary-Window Reserve Vignette and Benchmark Comparison

At a shortage-to-reserve cost ratio $\rho = 10$ on the transition boundary band, the boundary-router gate-bin policy costs 95.13M, outperforming the same-model global rule (99.55M), while the low-RMSE Graph WaveNet with physical-bin reference achieves 84.31M. This translates into approximately 637 avoided MWh-equivalent shortage cells and a 442k illustrative reserve-cost-scale marker at 100 EUR/MWh.

G. Point of Common Coupling (PCC) Portfolio Smoothing & Decadal Durability (Branch F)

Crucially, to verify reserve benefits survive spatial smoothing, we evaluate aggregate power at the Point of Common Coupling (PCC) bus ($P_{\text{farm}} = \sum_{i \in \mathcal{F}_t} P_{i,t}$, averaging ~ 121 active turbines). Across the full operational envelope, joint posterior aggregate quantile pricing reduces reserve cost by −11.22M kWh (95% bootstrap CI [−21.55M, −2.85M], strictly excluding zero) against global PCC quantiles and by −14.94M kWh (CI [−27.06M, −6.23M]) against Gaussian parametric sizing; in transitional regimes (10%–90% pitching), it saves −1.48M kWh (CI [−2.14M, −1.07M]) over continuous physical pitch, confirming economic value after fleet-wide error cancellation.

Across two European commercial wind plants spanning 17.6 cumulative operating years (Kelmarsh, 9 years, 2016–2024, 6 MM92 turbines; Penmanshiel, 8.6 years, 2016–2024, 15 MM82 turbines), neural backbones frozen at commissioning

paired with rolling two-year quantile recalibration recover 0.804M kWh (804,000 kWh) of annual drift loss per site. In full-sample testing across all 13 chronological rolling folds, 12 out of 13 folds demonstrate cost savings, achieving exact binomial sign test significance $p = 0.00171$. Cumulative walk-forward pooled savings reach −6.19M kWh on Kelmarsh (95% CI [−7.52M, −4.77M]) and −4.34M kWh on Penmanshiel mature folds (95% CI [−8.24M, −0.44M]), establishing the multi-year durability of the frozen-backbone framework.

VI. DISCUSSION

A. Accuracy, accountability and deployment gates

The accuracy cost is not secondary: iTransformer, Graph WaveNet, and lag baselines remain superior whole-sample forecasters on WTB; the routed model should pair with an established low-RMSE forecaster when aggregate accuracy is paramount. Its targeted role is an accountable operating-state diagnostic for boundary-specific decisions. External testing is graded: La Haute Borne demonstrates anchor-observable replication (five-seed routing NMI 0.941, ARI 0.971; canonical NMI 0.975; withheld probe 0.674/0.575). On Kelmarsh the probe reaches 0.340/0.378, and on Penmanshiel 0.302 (signature_core 0.195, far above permuted controls; Supplementary Table A9d). Multi-channel consequence quality governs boundary recoverability across heterogeneous turbine models.

Crucially, external walk-forward rolling evaluation establishes empirical physical boundary conditions for the pre-registered admission protocol (Supplementary Table A14). On EN-GIE La Haute Borne (4 turbines, 99% complete pitch), walk-forward pooled quarterly evaluation (Q1–Q3) incurs a positive cost delta of +1.01M kWh (95% bootstrap CI [+0.29M, +1.76M], strictly excluding zero) versus the global quantile and +0.92M kWh (CI [+0.26M, +1.61M]) versus continuous physical pitch (Supplementary Table A11h). This reveals quantile variance amplification under acute seasonal drift: on a miniature 4-turbine site without Point of Common Coupling (PCC) spatial portfolio smoothing, short quarterly slices suffer variance spikes during autumn regime shifts (+968.1k kWh in Q3, while Q1–Q2 cross zero). Expanding calibration to a 180-day annual window compresses the cost gap to +43k kWh (CI [−28.7k, +141.2k], strictly crossing zero), bounding soft-posterior reserves to PCC-smoothed plants and degraded/pitch-sparse telemetry.

Regarding decadal wear and multi-year climate cycles without online retraining, Section-V-G demonstrated that “Quantile Recalibration under Frozen Backbone” recovers 0.804M kWh of annual drift loss per site across 17.6 cumulative operating years ($p = 0.00171$). While static frozen backbones suffer multi-year drift (losses of −2.24M and −13.83M), two-year rolling recalibration yields cumulative walk-forward pooled savings of −6.19M kWh on Kelmarsh and −4.34M kWh on Penmanshiel mature folds (Supplementary Table A11g), with 10 of 11 mature annual folds strictly excluding zero.

Furthermore, evaluating across shortage penalty ratios $\rho \in \{5, 10, 20\}$ articulates an operational envelope (“Telemetry Availability $\times$ Penalty Ladder”). At $\rho = 10$, the soft gate dominates on both commercial farms. At $\rho = 20$, asymmetry emerges: at Kelmarsh, physical rules recover and surpass

soft-gate pricing by +0.95M kWh (CI [+0.15M, +1.65M]); at Penmanshiel, the soft gate leads physical rules by -7.51M kWh (CI [-13.51M, -1.25M]) while crossing zero versus the unconditioned global baseline (-5.15M kWh, CI [-12.47M, +2.17M]). Direct reserve deployment at new farms requires pitch or proxy observability, boundary support, compatible geometry, local recalibration, and a held-out routing pass (cross-farm held-out NMI reaches 0.557; Supplementary Table A14) [4].

VII. LIMITATIONS

WTB pseudo-labels derive from wind speed and pitch angle, channels present in full gate anchors. Counterfactual withheld-channel stress testing proves the boundary survives without direct pitch observations via consequence signatures across four farms (permuted controls at chance). Removing pitch dispersion retains non-power NMI 0.309 on WTB and 0.578 on La Haute Borne, confirming dispersion does not carry the signal. Yet routing remains an anchor-constrained diagnostic rather than anchor-free discovery [11], [15]. Inputs may contain history $\text{Patv}_{t-H+1:t}$ and issue-time Patv_t , while targets begin at $t + 1$. Expert semantics reflect the declared mapping; unassigned logits are not universal physical states.

Crucially, our evaluation is formally bounded to Level 1 local pre-dispatch risk screening proxies (Penalized Reserve-Shortfall Energy Index, PSREI at $\rho = 10$), designed to minimize unhedged imbalance entering real-time balancing. This local pre-dispatch screening explicitly abstracts away Level 2 transmission-level operations, including Alternating Current Optimal Power Flow (AC-OPF), network thermal and voltage constraints, security-constrained unit commitment (SCUC), and multi-stage financial cashflows or dynamic locational marginal pricing (LMP) [21], [23]. The PSREI proxy should be viewed as an upstream risk filter feeding into, rather than replacing, ISO/TSO physical market clearing. Strongest claims remain five-seed WTB recovery, mechanism intervention, La Haute Borne replication, and same-router reserve triage.

VIII. CONCLUSION

This study formalizes and validates a defense-in-depth architecture combining high-fidelity aerodynamic physical priors with an autonomous spatio-temporal dynamic graph blind-spot fallback for wind plant operational reserve pricing. Across a 5-seed multi-regime benchmark on the 134-turbine WTB plant, we established that while deterministic aerodynamic rules achieve optimal efficiency under clean telemetry (871,408 kW · h at $h = 6$), they suffer catastrophic brittleness under communication delays (>12.5% violation rates, breaching compliance limits). Conversely, shallow GBDT baselines collapse across dispatch horizons ($h = 6$, inflating costs by 13.44% to 29.15%, $p < 0.0001$), and uncalibrated two-stage direct pinball regression over-estimates reserves by up to 72.02% ($p < 0.01$).

The proposed jointly-learned spatio-temporal graph resolves this operational dilemma. By dynamically extracting electromechanical consequence signatures (power transients, voltage/reactive dynamics) when primary pitch telemetry is delayed or unobservable, the deep fallback maintains strict safety compliance (<8.1% violation rate) while delivering the lowest degraded reserve cost (1,149,897 kW · h at $h = 6$). Dynamic MoE routing provides critical fault isolation under continuous telemetry impairment ($p = 0.0251$ under 6-step delay and

$p < 0.05$ under noise). Fleet-wide Point of Common Coupling (PCC) spatial portfolio smoothing preserves an economic savings of -11.22M kWh (95% CI [-21.55M, -2.85M]) over global PCC baselines. Decadal walk-forward evaluation across 17.6 commercial operating years confirms rolling recalibration under frozen backbones recovers 0.804M kWh/year ($p = 0.00171$). The framework establishes an auditable, physics-data synergistic reserve pricing layer for digitalized power grids.

IX. AI USE STATEMENT

The authors used OpenAI ChatGPT/Codex only for language editing, consistency checks, and submission-material drafting; all data, analyses, references, conclusions, and final text were reviewed and controlled by the authors.

X. CODE AND DATA AVAILABILITY

The raw KDD Cup 2022, ENGIE La Haute Borne, Kelmars, and Penmanshiel SCADA datasets are public (raw third-party data not redistributed). Code, configurations, releasable derived tables, figure data, and checkpoints will be made available with the article. The reproduction package covers WTB routing, reserve audit, anchor-stress caches, early-warning label degradation, classifier control, class-weight sensitivity, external diagnostics, and evidence-freeze protocols across declared seeds. Results are statistically reproducible across declared seeds rather than bitwise deterministic across all GPU/CUDA environments. The analysis involves no human subjects.

REFERENCES

- [1] P. Pinson, "Wind energy: Forecasting challenges for operational management," *Statistical Science*, vol. 28, no. 4, pp. 564–585, 2013.
- [2] R. Doherty and M. O'Malley, "A new approach to quantify reserve demand in systems with significant installed wind capacity," *IEEE Transactions on Power Systems*, vol. 20, no. 2, pp. 587–595, 2005.
- [3] Y. Wang, F. Zhang, H. Kou, R. Zou, Q. Hu, and J. Wang, "A review of predictive uncertainty modeling techniques and evaluation metrics in probabilistic wind speed and wind power forecasting," *Applied Energy*, vol. 396, p. 126234, 2025.
- [4] J. Tautz-Weinert and S. J. Watson, "Using SCADA data for wind turbine condition monitoring: A review," *IET Renewable Power Generation*, vol. 11, no. 4, pp. 382–394, 2017.
- [5] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang, "Graph WaveNet for deep spatial-temporal graph modeling," in *Proceedings of the 28th international joint conference on artificial intelligence*, 2019, pp. 1907–1913.
- [6] S. Guo, Y. Lin, N. Feng, C. Song, and H. Wan, "Attention based spatial-temporal graph convolutional networks for traffic flow forecasting," in *Proceedings of the AAAI conference on artificial intelligence*, 2019, pp. 922–929.
- [7] L. Bai, L. Yao, C. Li, X. Wang, and C. Wang, "Adaptive graph convolutional recurrent network for traffic forecasting," in *Advances in neural information processing systems*, 2020, pp. 17804–17815.
- [8] J. Park and J. Park, "Physics-induced graph neural network: An application to wind-farm power estimation," *Energy*, vol. 187, p. 115883, 2019.
- [9] D. Kim, G. Ryu, and C. Moon, "Accuracy of a short-term wind power forecasting model based on deep learning using LiDAR-SCADA integration: A case study of the 400-MW Anholt offshore wind farm," *Applied Energy*, vol. 373, p. 123882, 2024.
- [10] S. Daenens, T. Verstraeten, P.-J. Daems, A. Nowé, and J. Helsen, "Spatio-temporal graph neural networks for power prediction in offshore wind farms using SCADA data," *Wind Energy Science*, vol. 10, pp. 1137–1152, 2025.
- [11] N. Zehabiyen-Rezaie, A. Iosifidis, and M. Abkar, "Physics-guided machine learning for wind-farm power prediction: Toward interpretability and generalizability," *PRX Energy*, vol. 2, no. 1, p. 013009, 2023.
- [12] N. Shazeer et al., "Outrageously large neural networks: The sparsely-gated mixture of experts layer," in *International conference on learning representations*, 2017.
- [13] W. Fedus, B. Zoph, and N. Shazeer, "Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity," *Journal of Machine Learning Research*, vol. 23, no. 120, pp. 1–39, 2022.
- [14] A. Karpatne et al., "Theory-guided data science: A new paradigm for scientific discovery from data," *IEEE Transactions on Knowledge and Data Engineering*, vol. 29, no. 10, pp. 2318–2331, 2017.
- [15] G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, and L. Yang, "Physics-informed machine learning," *Nature Reviews Physics*, vol. 3, no. 6, pp. 422–440, 2021.
- [16] E. A. Bossanyi, "The design of closed loop controllers for wind turbines," *Wind Energy*, vol. 3, no. 3, pp. 149–163, 2000.
- [17] F. Bianchi, H. De Battista, and R. J. Mantz, *Wind turbine control systems: Principles, modelling and gain scheduling design*. London: Springer, 2006.
- [18] L. Y. Pao and K. E. Johnson, "Control of wind turbines: Approaches, challenges, and recent developments," *IEEE Control Systems Magazine*, vol. 31, no. 2, pp. 44–62, 2011.
- [19] K. E. Johnson, L. J. Fingersh, M. J. Balas, and L. Y. Pao, "Methods for increasing Region 2 power capture on a variable-speed wind turbine," *Journal of Solar Energy Engineering*, vol. 126, no. 4, pp. 1092–1100, 2004.
- [20] E. Ela, M. Milligan, and B. Kirby, "Operating reserves and variable generation," National Renewable Energy Laboratory, Golden, CO, NREL/TP-5500-51978, 2011.
- [21] J. B. Bremnes, "Probabilistic wind power forecasts using local quantile regression," *Wind Energy*, vol. 7, no. 1, pp. 47–54, 2004.
- [22] Y. Zhang, J. Wang, and X. Wang, "Review on probabilistic forecasting of wind power generation," *Renewable and Sustainable Energy Reviews*, vol. 32, pp. 255–270, 2014.
- [23] Z. Zhou et al., "Application of probabilistic wind power forecasting in electricity markets," *Wind Energy*, vol. 16, no. 3, pp. 321–338, 2013.
- [24] Y. Li, R. Yu, C. Shababi, and Y. Liu, "Diffusion convolutional recurrent neural network: Data-driven traffic forecasting," in *International conference on learning representations*, 2018.
- [25] H. Hersbach et al., "The ERA5 global reanalysis," *Quarterly Journal of the Royal Meteorological Society*, vol. 146, no. 730, pp. 1999–2049, 2020.
- [26] H. A. Nielsen, L. Madsen, and T. S. Nielsen, "Using quantile regression to extend an existing wind power forecasting system with probabilistic forecasts," *Wind Energy*, vol. 9, no. 1–2, pp. 95–108, 2006.
- [27] J. Zhou et al., "SDWPF: A dataset for spatial dynamic wind power forecasting over a large turbine array," *Scientific Data*, vol. 11, p. 649, 2024.
- [28] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, "A time series is worth 64 words: Long-term forecasting with transformers," in *International conference on learning representations*, 2023.
- [29] Y. Liu et al., "iTransformer: Inverted transformers are effective for time series forecasting," in *International conference on learning representations*, 2024.
- [30] A. Das, W. Kong, A. Leach, and R. Sen, "Long-term forecasting with TiDE: Time-series dense encoder," in *Transactions on machine learning research*, 2023.